In [ ]:
```
# Python ka Chilla with Baba Ammar
## How to use jupyter Note book
### Basics of python
1- My first program

2- My second program
```

In [6]:
```
My first programming in python
print("AI ka chilla with BABA GEEE")

print("hello world")
print("learning pthon with Dr.Ammar Tufail Faisalabadi with jugtojugti")
```

```
AI ka chilla with BABA GEEE
hello world
learning pthon with Dr.Ammar Tufail Faisalabadi with jugtojugti
```

In [ ]:
```
**02- Operators**
```

In [8]:
```
print(2+3)
print(3-1)
print(6/2)
print(2*3)
print(13%2)
print(6//2)
print(2**2)

print(3**2/2*3/3+6-4)
```

```
5
2
3.0
6
1
3
4
6.5
```

In [ ]:
```
**03- Strings**
```

In [9]:
```
print("hello world")
print("we are learning python with Ammar Tufail")
print('test for a single quote')
print("test for double quotes")
print('''test for double quotes''')
print("What's up")
#Srings clear.
```

```
hello world
we are learning python with Ammar Tufail
test for a single quote
test for double quotes
test for double quotes
What's up
```

04- comments

In [10]:
```
#print("How are you") it can be used for comment out. or ctrl + /, now it will not
print("we are learning python with Ammar") #print a string
print(2+4) # print operatro function with number
```

```
we are learning pthon with Ammar
6
```

In [ ]:
```
***05- Variables***
```

In [11]:
```
print("Hello world")
# Variables objects containing specific values.
x = 5
print(x)
y = "we are learing pyhon with ammar" #string variable
print(y)

x = 15 # it can be changed as 5+10 in both ways
print(x)


#types of varabiles.
type(x)
print(type(x))


print(type(y))

#print_type_variable
#rules to assign a variable
#1 The variable should contian letters, numbers. and underscores
#2 Do not start with numbers
#3 spaces are not allowed.
#4 Do not use keywords in function like break, mean, median test and etc
#5 Short and descriptive
#6 case sensitivity, try to use lower case


fruit_basket= "mangoes, oranges"
print(fruit_basket)

fruit_basket= 8
fruit_basket= 15
fruit_basket= "mangoes"
print(fruit_basket)

# as line number increased then it will increased and shoed the last line command.
```

```
Hello world
5
we are learing pyhon with ammar
15
<class 'int'>
<class 'str'>
mangoes, oranges
mangoes
```

In [ ]:
```
***06- Input Variables***
```

In [12]:
```
#print("hello world")

#fruit_basket= "mangoes"
#print(fruit_basket)

#input function
#fruit_basket=input("what is your favourite fruit?")
#print(fruit_basket)

#it means if we ask the quesions here then and run this command, then in the termin

#input function of sencond stage.
#name= input("what is your name? ")
#greetings="hello"

#print(greetings, name)

#another way of stage 2 function
#name= input("what is your name ?")
#print("hello!", name)

#3rd stage input function
name= input ("what is your name? ")
age= input ("How old are you? ")
greetings="Hello!"

print(greetings, name, "You are still young")

#input _aamar_you are still young
```

```
Hello! ahmad You are still young
```

In [ ]:
```
***07- Conditional logics***
```

In [13]:
```
#logical operators are either "true or false, or "yes or no" or "1 or 0"
# equal to                              ==
# not equal to                          !=
# less than                             <
# greater than                           >
# less than and equal to                <=
# greater than and equal to               >=

#is 4 equal to 4?
print(4==4)
print(4!=4)
```

```
print(4>3)
print(3>6)
print(3<=6)
print(5>=4)

#applicaitons of logical operators
#ahamd_age=4
#age_at_school=5
#print(ahmad_age==age_at_school)


#input operator and logicals
age_at_school=5
ahmad_age=input("How old is ahmad? ") #input function
ahmad_age=int(ahmad_age) #if result show false result with string in termainal then
print(type(ahmad_age))
print(ahmad_age==age_at_school) #logical operator

#convert input
```

```
True
False
True
False
True
True
<class 'int'>
True
```

In [ ]:

```
***08- Conversion***
```

In [14]:

```

```

```
True
False
True
False
True
True
<class 'int'>
True
```

In [ ]:

```
***08- Conversion***
```

In [15]:

```
x = 10  # integer
y = 10.2  # float
z = "Hello" #string

#implicit type conversion
#x= x+y # if we swap values like in decimal then it will show float class. is alway

#print(x, "type of x is:", type(x))


#explicit type conversion
#age=input("what is your age? ")
```

```
#age=int(age) #  an other way or alternative way for conversion of string to intege

#print(type(int(age)))

#name
name=input("what is your name? ")
print(name, type(str(name)))

#type_coversion
```

```
ahmad <class 'str'>
```

In [ ]:
```
***09- if_else_elif***
```

In [16]:
```
required_age_at_school = 10
ahmad_age = 2
# question: can ahmad go to school ?

if ahmad_age==required_age_at_school:
    print("congratulation: ahmad can join the school")
elif ahmad_age > required_age_at_school:
   print("ahmad should join higher secondary school")
elif ahmad_age == 2:
    print("you should take care of him, he is still a baby!")
else:
    print("ahmad can not go to school or ahmad join higher to school")
```

```
you should take care of him, he is still a baby!
```

In [ ]:
```
***10- Functions
```

In [17]:
```
#defining a Functions
# 1
#def print_codanics():
    #print("we are learning with Ammar")
#print("we are learning with Ammar")
#print("we are learning with Ammar")
#print("we are learning with Ammar")

#print_codanics()


# 2
#def print_codanics():
#text = "we are learning with Ammar"
#print(text)
#print(text)
#print(text)
#print_codanics()


# 3
#def print_codanics(text):
    #print(text)
    #print(text)
```

```
    #print(text)

#print_codanics("we are learning python with Ammar")

# 4 defing a function with if, alif, and else statements

def school_calculator(age, text):
    if age==5:
        print("ahmad can join the school")
    elif age>5:
        print("ahmad should go to higer school")
    else:
        print("ahmad is still a baby")

school_calculator(15, "ahmad")


#Fucture Function
def future_age(age):
    new_age=age+20
    return new_age
    print(new_age)
future_predicted_age= future_age(5)
print(future_predicted_age)

#i leanrt the function. it is also a small machine learning model. i understand fun
```

```
ahmad should go to higer school
25
```

In [ ]:

```
***11- loops***
```

In [18]:

```
#while and for loops
#while loops, To do work again and again like a baby to send him school daily by hi


#x=0
#while (x<5):
    #print(x)
    #x=x+1

# For loop
#for x in range(5,10):
    #print(x)


# array
days = ["mon", "tue", "wed", "thu", "fri", "sat", "sun"]

for d in days:
    #if (d=="fri"):break #loop stops # loop is not working.
    if (d=="fri"): continue # here friday is missing but loop continue forward.
    print(d)
```

```
mon
tue
wed
thu
sat
sun
```

In [19]:
```
#if you want to print tyhe value of pi

import math
print("the value of pi is ", math.pi)

import statistics
x= [150,250,350,450]

print(statistics.mean(x)) #we can also used readymade liabraies in python, like mea

#name of important liabrieres in python like numpy, pandas etc
```

```
the value of pi is  3.141592653589793
300
```

In [ ]:
```
***12_ import liabraries***
```

In [20]:
```
#if you want to print tyhe value of pi

import math
print("the value of pi is ", math.pi)

import statistics
x= [150,250,350,450]

print(statistics.mean(x)) #we can also used readymade liabraies in python, like mea

#name of important liabrieres in python like numpy, pandas etc
```

```
the value of pi is  3.141592653589793
300
```

In [ ]:
```
***13- Troubleshooting***
```

In [21]:
```
#print("we are learning python wiht ammar")

#print(25/0) #runtime error

name= "ammar"
print("hello", name) #sematic error. its more difficult becase it yours your person
# we use the plus button + for joint two words. like shahidbutt but in command in s
```

```
hello ammar
```